PAPER_704: Horsehead Nebula (Barnard 33): Infrared Erosion UQFF Analysis

Class: `HorseheadNebulaBarnard33UQFF` **CP4 Entry:** #288 **Keywords:** Horsehead Nebula, Barnard 33, dark nebula, radiation pressure, erosion, Sigma Orionis, UQFF **Session:** 175 | **Version:** v5.32 **Source:** grok_share_ba508f76c8e.txt

Abstract

The Horsehead Nebula (Barnard 33) is a dark molecular cloud in Orion at 1500 ly, photographically famous as a silhouette against the emission nebula IC 434. It is slowly being photoevaporated by ultraviolet radiation from Sigma Orionis ($L \approx 10^5 L_\odot$). The UQFF MUGE is derived for this system incorporating radiation pressure erosion, gravitational self-collapse, and EM aether coupling.

1. System Parameters

Parameter	Value	Description
M_{neb}	$120 M_\odot$	Total mass (100 gas + 20 YSOs)
r_{neb}	1.182×10^{16} m	Half-span (1.25 ly)
L_{Sigma}	$10^5 L_\odot$	Sigma Orionis luminosity
ρ_{neb}	10^{-21} kg/m^3	Nebula density
τ_{erode}	1.578×10^{14} s	Erosion timescale (5 Myr)
E_0	0.2	Fractional erosion rate

2. Master UQFF Gravity Equation

$$g_{HH}(t) = \frac{GM}{r^2}(1 + H_z t)(1 - E(t))(1 + f_{TRZ}) + P_{rad} + q(v \times B)\left(1 + \frac{\rho_{UA}}{\rho_{SCm}}\right) \times 10^{-12}$$

2.1 Erosion Term

$$E(t) = E_0\left(1 - e^{-t/\tau_{erode}}\right)$$

At $t = 1$ Myr: $E \approx 0.039$, so $(1 - E) \approx 0.961$.

2.2 Radiation Pressure Acceleration

$$P_{rad} = \frac{L_{star}}{4\pi r_{neb}^2 c} \cdot \frac{\rho_{neb}}{m_H} \approx 4.35 \times 10^{-5}\,\text{m/s}^2$$

This represents the net UV ram pressure eroding the nebula pillar.

3. Numerical Result

$$g_{HH}(t = 1\ \text{Myr}) \approx 1.097 \times 10^{-3}\,\text{m/s}^2$$

4. Infrared Observation

Hubble Space Telescope WFC3 IR imaging (2013) reveals the nebula in the NIR, with proto-stellar objects embedded in the pillar tip. The UQFF framework predicts the gravitational collapse rate consistent with the 100 kyr protostellar formation timescale.

References

- Reipurth & Bally (2001), ARA&A, 39, 403
- Pound (1998), Astrophys. J., 493, L113
- UQFF Framework, Star-Magic Session 175

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